

AGGRIYA ANAND

AI & Machine Learning Engineer | Generative AI

 Lucknow, UP, India  anand.bietlko@gmail.com  +91 8299839096  linkedin.com/in/aggriya-anand-25241938a
 github.com/aggriya0007

PROFESSIONAL SUMMARY

Dynamic and results-driven Artificial Intelligence & Machine Learning engineer with expertise in predictive modeling, Generative AI, computer vision, and IoT hardware integration. Experienced in delivering practical engineering solutions—ranging from intelligent healthcare diagnostic tools and real-time OpenCV driver safety systems integrated with Arduino sensors to autonomous IoT firefighting robotics. Proven ability across onsite and virtual industry internships to bridge hardware-software boundaries and deploy robust data-driven applications.

INTERNSHIP & EXPERIENCE

AIML Intern **ONSITE** **June – August**
Skill Manthan Technologies Private Limited *Lucknow, UP*

- Completed intensive onsite training and practical development in core Artificial Intelligence and Machine Learning methodologies.
- Designed, trained, and fine-tuned predictive machine learning models using Python and Scikit-Learn.
- Developed scalable data preprocessing workflows, feature engineering methods, and model evaluation routines for complex datasets.
- Collaborated with cross-functional technical teams to integrate ML models into database-driven software architectures.

Generative AI Intern (IBM PBEL) **VIRTUAL** **June – August**
IBM Project Based Experiential Learning *Virtual*

- Gained hands-on experience in Generative AI architectures, Large Language Models (LLMs), Prompt Engineering, and RAG frameworks.
- Explored state-of-the-art model deployment strategies, multi-modal applications, and automated document synthesis.
- Implemented project modules under IBM mentorship, evaluating generative outputs and optimizing model efficiency.
- Earned certification for successful project execution and technical evaluations in the IBM PBEL program.

KEY PROJECTS

1. AI Healthcare Diagnosis Assistant Application **Project Lead**

- Engineered an intelligent diagnostic application utilizing machine learning models to analyze clinical parameters and assist medical risk assessment.
- Implemented robust database management for tracking structured patient records and historical data securely.
- Built an automated report generation system, formatting diagnostic summaries and graphical insights dynamically.

2. Driver Drowsiness Detection System **Hardware & Software Integration**

- Developed a computer vision system utilizing OpenCV to detect driver fatigue by computing Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) in real-time.
- Seamlessly integrated an Arduino microcontroller with an external blink sensor to create a hybrid hardware-software driver monitoring framework.
- Triggered instant audible and physical safety alerts upon detecting continuous eye closure or yawning micro-sleep events.

3. IoT Firefighting Robot **IoT & Embedded Systems**

- Designed and built a fully functional autonomous firefighting robot powered by an Arduino board and multi-module IoT sensors.
- Integrated flame and thermal sensors alongside obstacle-avoidance modules for autonomous navigation and immediate fire detection.
- Configured motor driver modules and automated water-pump actuators for remote or autonomous fire suppression capabilities.

TECHNICAL SKILLS

AI / ML & GenAI:	Machine Learning, Deep Learning, Generative AI, LLMs, RAG, Prompt Engineering, Neural Networks, Computer Vision
Languages & Libs:	Python, SQL, HTML/CSS, Scikit-Learn, Pandas, NumPy, OpenCV
Tools & Microcontrollers:	MySQL, Git/GitHub, VS Code, Arduino, IoT Hardware Modules

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering (AI&ML) <i>BIET Lko (Bansal Institute of Engineering and Technology)</i>	2023 – 2027 <i>Lucknow, UP</i>
Higher Secondary Education (10+2 - ISC) <i>SKD Academy (ISC Branch)</i>	Completed <i>Lucknow, UP</i>